

Twisha Patel

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Summary

Results-driven **AI/ML Engineer** with hands-on experience building and deploying production-grade Computer Vision, NLP, and LLM systems. Designed end-to-end ML pipelines spanning data ingestion, training, evaluation, deployment, and monitoring. Strong in Python and modern ML stacks (PyTorch, TensorFlow, OpenCV, Hugging Face, FastAPI, Docker). Proven track record of improving model performance, reducing latency, and delivering scalable, real-world AI solutions, including production-ready RAG systems in open source.

Education

Indian Institute of Technology, Madras *Dec 2025 – Jun 2026*
Advanced PG in Agentic AI Workflows and Agentic Systems Development

ITM SLS Baroda University, Vadodara, Gujarat *Aug 2021 – May 2025*
B.Tech in Computer Science Engineering — CGPA: 8.6/10

Shri J.R Shah Bright English Medium School, Vadodara, Gujarat *June 2020 – July 2021*
Higher Secondary School — Grade: B2

Experience

Software Engineer *April 2026 – Present*
Artem HealthTech, Ahmedabad, India

- Engineered LIS (Laboratory Information System) integration pipelines by implementing data synchronization logic for machine-generated data, ensuring reliable ingestion, validation, and storage within the platform.
- Designed and optimized healthcare data pipelines for ingestion, validation, transformation, and reporting. Built dashboards and automated workflows to improve operational efficiency and data-driven decision-making.

Jr. Software Engineer (AI/ML) *July 2025 – March 2026*
Sundaram Technologies, Vadodara, India

- Designed and deployed industrial computer vision systems that increased detection reliability (F1-score) while cutting preprocessing latency by **35%**.
- Optimized and productionized ML models using TensorFlow, PyTorch, and OpenCV; containerized services with Docker and CI/CD, reducing release cycles from **14 days to 3 days**.
- Built scalable inference pipelines and monitoring workflows to ensure model stability in production environments.

AI/ML Intern *Jan 2025 – July 2025*
Artem Health Tech, Ahmedabad, India

- Engineered a machine-integration data pipeline that automated device data synchronization, reducing manual effort by **40%** and improving data freshness from **daily to hourly**.
- Implemented real-time object detection using MMDetection (RTMDet-tiny) on live webcam streams at **25–30 FPS** with sub-**50 ms** preprocessing latency.
- Developed a speech-to-text system using open-source models with agentic workflows, achieving **12–15% WER** on internal benchmarks.
- Fine-tuned a finance-focused LLM (Flan-T5), improving exact-match accuracy for domain-specific Q&A by **18%**.

Technical Skills

Languages: Python, JavaScript, SQL, PHP, Bash

ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Hugging Face, Transformers

LLMs & NLP: Fine-tuning, Prompt Engineering, RAG, Embeddings, Evaluation, Deployment

MLOps & Deployment: Docker, FastAPI, CI/CD, GitHub Actions, Model Monitoring

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, EDA, Reporting (CSV, PDF, JSON)

Databases: MySQL, SQLite, CouchDB

Web: FastAPI, REST APIs, Streamlit, HTML, CSS, Bootstrap

Key Projects

Pneumonia Detector AI

- Built an explainable AI system for pneumonia detection using EfficientNet (PyTorch) with GradCAM visualizations for clinical interpretability.
- Implemented a human-in-the-loop workflow with automated retraining, quality checks, and session logging to continuously improve model performance.
- Delivered a privacy-first system supporting batch analysis and downloadable reports (CSV, PDF, JSON, FHIR), packaged in Docker for easy deployment.

AutoML Pipeline Service

- Architected and deployed an end-to-end AutoML platform (Streamlit + FastAPI) for dataset profiling, model selection, training, inference, and logging.
- Implemented reproducible experiment tracking with configurable retention policies and benchmark comparisons.

Real-Time Object Detection (MMDetection)

- Built a real-time detection system using RTMDet-tiny with OpenCV for live video processing.
- Visualized predictions with bounding boxes and class labels using COCO-pretrained models.

Traffic Sign Recognition

- Developed and trained a CNN using TensorFlow/Keras on the GTSRB dataset with extensive augmentation.
- Applied advanced OpenCV preprocessing techniques to improve generalization and robustness.

Finance Domain Small LLM

- Fine-tuned Flan-T5 on curated financial datasets for domain-specific Q&A.
- Designed evaluation and prompt strategies to improve contextual accuracy and relevance.

Open Source Contributions

LlamaIndex — Open Source Contributor

2026

- Designed and contributed a production-ready FastAPI-based RAG example using LlamaIndex with fully local Ollama models.
- Built an offline document ingestion and vector search pipeline using local embeddings (no API keys required).
- Integrated API endpoints for querying indexed documents and ensured quality via CI, linting, and maintainer reviews.
- Collaborated with maintainers on architecture, model selection, and best practices; contribution successfully merged.